

Network Perimeter Defense Lab

PROJECT BRIEF

DoS Attack Simulation & Real-Time Firewall Mitigation using pfSense + Kali Linux + Ubuntu

TYPE	Home Lab / Cybersecurity Project	ENVIRONMENT	Oracle VirtualBox (3 VMs)
DOMAIN	Network Security · Firewall Engineering · Threat Emulation	DURATION	Self-Directed Lab Build

PROBLEM · ACTION · RESULT

PROBLEM

A permissive firewall ruleset allowed an external attacker (Kali Linux) to launch an unmitigated SYN/ICMP flood against an internal Ubuntu host. The flood generated hundreds of thousands of malicious packets, reducing the victim's internet connectivity by over 77% and producing a measurable Denial-of-Service condition — verified live in Wireshark.

ACTION

Architected a three-VM isolated lab environment in VirtualBox. Engineered pfSense firewall rules to expose and then defend the attack surface. Executed volumetric flood attacks using hping3 in both SYN and ICMP flood modes. Captured and analysed all traffic in Wireshark to confirm attack signatures. Designed and inserted a WAN-interface block rule at the top of the pfSense ruleset with logging enabled to intercept all malicious traffic before permissive rules could fire.

RESULT

The block rule reduced attacker-sourced packets reaching the victim to zero. hping3 reported 100% packet loss on the attacker side. Wireshark on Ubuntu showed immediate traffic normalisation. pfSense firewall logs provided an auditable, timestamped record of every blocked packet — demonstrating complete mitigation of the attack with full forensic evidence of the block.

KEY METRICS

236,492+	SYN Packets Transmitted (flood)	100%	Packet Loss on Attacker Post-Rule	77%+	Victim Internet Loss (Pre-Mitigation)	0	Flood Packets Reaching Target
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TECHNICAL DETAILS

Attack Methodology

Configured Kali Linux (192.168.1.22) on the WAN-bridged segment and injected a static route to reach the isolated LAN subnet (192.168.2.0/24) via pfSense's WAN IP. Launched two flood variants:

- **SYN Flood:** hping3 -S -p 80 --flood 192.168.2.100 — targeted TCP port 80, exhausting connection state on the victim
- **ICMP Flood:** hping3 -I --flood 192.168.2.100 — volumetric ICMP echo request storm at maximum transmission rate

Firewall Mitigation

Navigated to pfSense WebGUI → Firewall → Rules → WAN and engineered a Block rule positioned above all permissive rules. Rule order in pfSense is deterministic (first match wins) — placing the block rule at the top ensured flood traffic was intercepted before any pass rules could evaluate it. Logging was enabled on the block rule to generate an auditable record of all dropped packets.

Verification Approach

Post-mitigation efficacy was confirmed at three independent observation points simultaneously:

- Attacker (Kali): hping3 statistics — 100% packet loss confirmed
- Victim (Ubuntu): Wireshark — SYN flood packets ceased immediately
- Firewall (pfSense): Log entries confirmed each blocked packet with source IP, destination, and timestamp

Skills Demonstrated

- Firewall Rule Engineering
- Stateful Packet Filtering
- DoS/DDoS Threat Emulation
- Network Traffic Analysis
- Wireshark Packet Inspection
- TCP/IP Protocol Analysis
- DHCP Scope Configuration
- Static Route Management
- Network Segmentation Design
- Attack Lifecycle Documentation
- Firewall Log Analysis
- Virtualisation Networking

TOOLS & TECHNOLOGY STACK

Category	Tool / Technology	Purpose
Firewall / Router	pfSense CE 2.7.x	Edge security, rule enforcement, DHCP, logging
Attack Platform	Kali Linux + hping3	DoS flood generation (SYN & ICMP modes)
Victim / Monitor	Ubuntu 22.04 Desktop	Target workstation, Wireshark capture point
Packet Analysis	Wireshark	Live packet capture, attack signature verification
Virtualisation	Oracle VirtualBox 7.x	VM lifecycle, network adapter configuration
Networking	Bridged + Internal Adapters	WAN/LAN segmentation, isolated LabNet

NEXT STEPS & EXTENSIONS

IDS/IPS Integration	Deploy Snort or Suricata on pfSense to add signature-based detection alongside reactive firewall rules
Rate Limiting	Implement pfSense traffic shaping to throttle volumetric floods without full blocks — modelling real-world graduated response
Multi-Source DDoS	Simulate distributed floods from multiple Kali instances to test rule scalability and GeoIP blocking

SIEM Integration

Forward pfSense syslog to Graylog or ELK Stack for centralised log analysis and alerting dashboards

Self-directed cybersecurity home lab project | Conducted in an isolated VirtualBox environment | For educational and portfolio purposes only